Aaryan Nakhat

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Experience

Founding AI/ML Engineer — Sep 2023 – Present
SalesAgents AI — *Mohali, India*

- Built **robust data pipelines** for data prep, cleaning, transformation, and synthetic data generation, along with **continuous evaluation and automated improvement pipelines**, reducing manual dependency by **75%**
- **Fine-tuned** multiple small-sized **LLMs** for diverse use cases including conversational AI, bidirectional translation, structured information extraction, topic modeling, action prediction, and custom guardrails
- **Preference-tuned fine-tuned LLMs** using techniques like **DPO** to better align model outputs with user preferences and improve response quality
- Iteratively developed a **RAG system** from scratch—starting with **vector similarity**-based retrieval (semantic/agentic chunking, re-ranking, query rewriting, HyDE), evolving to **Neo4j**-based knowledge graphs, and achieving best performance with **GraphRAG** (LightRAG)
- Developed and scaled **AI backend systems** using **LiveKit** with deep customization—implementing **multi-agent orchestration**, tool calling, caching strategies, load balancing, and observability for production-grade voice agents
- Optimized real-time voice agent infrastructure, reducing **on-call latency by 50%** and cutting operational **costs by 85%** through efficient system design and resource utilization

Software Development Intern — Jan 2023 – Sep 2023
ValueLabs — *Hyderabad, India*

- Developed a custom **fine-tuned insurance chat expert**, deployed on a large insurance company's website
- Designed **custom guardrails** on top of the fine-tuned LLM **for input and output control**
- Created an additional **auxiliary classifier** for guiding the LLM to its desired goal at each step

Education

ICFAI Foundation for Higher Education — Hyderabad, India
Bachelor of Technology in Data Science and Artificial Intelligence; CGPA: 9.78/10 — *Aug. 2019 – June 2023*

- Achievements: **gold medal** for all-round excellence (best student), **silver medal** for academic excellence, **100% scholarship in all 8 semesters**, won 2 intra-college coding competitions
- Extracurricular activities: vice president of the technical club, part of the university's cricket team
- Relevant Coursework: Probability, Statistics, Data Science, Machine Learning, Deep Learning, Data Visualization, Data Mining, Data Warehousing, Block Chain Technology

Projects

CricDex: Open Cricket Intelligence Platform *(demo) (code)* — May 2026 – Present

- Ingested cricket ball-by-ball data into **DuckDB**; modeled dismissal-aware, opponent-adjusted player skill via **Bayesian** inference (**JAX/NumPyro**, ADVI + NUTS) and scored each ball's match impact (Net Game Impact) with an **XGBoost** win-probability model (WPA-style per-delivery ΔWP)
- Engineered 12 novel context-adjusted metrics that go beyond single-number stats (batting average, economy...) to capture a player's true match impact
- Built a cross-league **Scout** (within-tier Bayesian skill-z look-alikes & replacement search across 6 competitions) and a real-rules IPL auction **Monte-Carlo**—editable retentions, overseas caps, franchise-personality bidding, relevance-weighted pricing

Transformer-Based Facial Emotion Recognition From Videos *(link)* — June 2023 – July 2023

- Engineered a **video-based emotion recognition system** by fine-tuning a **Vision Transformer** (ViT), tackling challenges of real-time processing and variable lighting to achieve a **top 1% F1 score** on the test set
- Fine-tuned a ViT on facial emotion recognition from images, and then extended it to video inputs

Certifications and Skills

Languages: Python, C, C++, SQL

Frameworks: PyTorch, LiveKit, LangChain, Vector databases, Neo4j, GraphRAG, Tableau, Power BI, FastAPI

Developer Tools: Linux, Git, VS Code, PyCharm, Jupyter Notebooks

Deployment: Docker, Kubernetes, vLLM, SGLang

Cloud Platforms: AWS, GCP, Azure

Certifications (clickable links): Python; Data Science and Machine Learning; Deep Learning; Custom Models, Layers and Loss Functions; Building LLM-Powered Applications; REST APIs; SQL